

Zurab Jashi

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EDUCATION

Perimeter Institute

Waterloo, ON

MSc in Physics

Relevant coursework: Quantum Field Theory, Quantum Information, Statistical Physics, String Theory, Mathematical Physics, Quantum Physics, Standard Model, Condensed Matter, Gravitational Physics

New York University

New York & Abu Dhabi

BSc in Mathematics, GPA: 3.94

Relevant coursework: Multivariable Calculus, Linear Algebra, Abstract Algebra, ODEs, PDEs, Chaos & Dynamics, Advanced Probability, Stochastic Calculus, Computational Physics, Topology, Differential Geometry, Theoretical Machine Learning, Applied Machine Learning

EXPERIENCE

Software Engineer – AI

Abu Dhabi, UAE

New York University

- Design and implement agent pipelines using LangGraph to help mathematicians generate mathematical ideas.
- Manage and optimize relational and vector databases, including PostgreSQL and pgVector.
- Parse, transform, and ingest large-scale scientific datasets into machine learning models using Python.
- Collaborate with mathematicians to develop better prompts that allow LLMs to search for novel ideas more effectively.
- Create UIs using FastAPI as a backend and React and Next.js for user interaction.

Software Engineer – Big Data

Tbilisi, Georgia

Bank of Georgia

- Developed Spark pipelines (batch and streaming) to calculate customer eligibility for cashback, discounts, and loyalty offers across 4M+ customers.
- Built real-time Kafka workflows that update customer eligibility instantly when data changes.
- Supported campaign launches by rerunning eligibility logic on demand for newly created offers.
- Handled large data volumes during events like Black Friday (up to 1.5B rows), improving reliability and performance.
- Worked with Java and QA engineers to investigate customer cases and fix logic issues in complex Spark code.
- Deployed updates from development to production using Git, Jenkins, Docker, and Kubernetes.

Software Engineering Intern – Machine Learning

Waterloo, Canada

Perimeter Institute for Theoretical Physics

- Implemented and tested the twinned regression algorithm, which restructures inputs from $x \rightarrow y$ to $(x_1, x_2) \rightarrow (y_1 - y_2)$, enabling ensemble-like behaviour and uncertainty estimation.
- Applied the method across neural networks, NN+kNN hybrids, and extensively on random forests.
- Ran large-scale experiments on the Graham supercomputer (Compute Canada); observed 5–10% accuracy gains on some datasets and underperformance on others, due to increased input dimensionality.
- Contributed to extending the method to a semi-supervised version; results published at arxiv.org/abs/2301.01383.

PROJECTS

Tutoring App | *Java, PostgreSQL, React, Docker*

- Building a tutoring platform with a team of developers to connect tutors and students from around the world.
- Developing a Java (Spring) backend and a React frontend with PostgreSQL as the main database.

Portfolio Website | *React, Next.js*

- React-based portfolio website deployed on Vercel.

AWARDS

Full Scholarship at *MSc Program at Perimeter Institute*

Silver Medal at *International Mathematics Competition*

Full Scholarship at *BSc Program at New York University*

Silver Medal at *International Physics Olympiad*

1st place at *Georgia's National Physics Olympiad*

Bronze Medal at *International Physics Olympiad*